Professional Ethics	2	2
Intellectual Property	1	1
Privacy and Civil Liberties	2	1
Communication	2	1
Sustainability	1	1
History	1	1
Economies of Computing	0	1
Security Policies, Laws, and Computer Crimes	2	1
Diversity, Equity, Inclusion and Accessibility	2	2
Total	**18**	**14**

Knowledge Units

SEP-Context: Social Context

Computers, the internet, and artificial intelligence - perhaps more than any other technologies - have transformed society over the past several decades, with dramatic increases in human productivity; an explosion of options for news, entertainment, and communication; and fundamental breakthroughs in almost every branch of science and engineering. It is imperative to recognize that this is not a one-way street. Society also affects computing, resulting in a complex socio-technical context that is constantly changing, requiring the perspective of history to put the present as well as the possible future into appropriate perspective.

Social Context provides the foundation for all other knowledge units in SEP, particularly Professional Ethics.

CS Core:

1. Social implications (e.g., political and cultural ideologies) in a hyper-networked world where the capabilities and impact of social media, artificial intelligence, and computing in general are rapidly evolving.
2. Impact of computing applications (e.g., social media, artificial intelligence applications) on individual well-being, and safety of all kinds (e.g., physical, emotional, economic).
3. Consequences of involving computing technologies, particularly artificial intelligence, biometric technologies, and algorithmic decision-making systems, in civic life (e.g., facial recognition technology, biometric tags, resource distribution algorithms, policing software) and how human agency and oversight is crucial.

4. How deficits in diversity and accessibility in computing affect society and what steps can be taken to improve equity in computing.

KA Core:

5. Growth and control of the internet, data, computing, and artificial intelligence
6. Often referred to as the digital divide, differences in access to digital technology resources and its resulting ramifications for gender, class, ethnicity, geography, and/or developing countries, including consideration of responsibility to those who might be less wealthy, under threat, or who would struggle to have their voices heard.
7. Accessibility issues, including legal requirements such as Web Content Accessibility Guidelines (www.w3.org/TR/WCAG21)
8. Context-aware computing

Illustrative Learning Outcomes:

CS Core:

1. Describe the different ways that computer technology (networks, mobile computing, artificial intelligence) mediates social interaction at the personal and collective levels.
2. Identify developers' assumptions and values embedded in hardware and software design, especially as they pertain to usability for diverse populations including under-served and those with disabilities.
3. Interpret the social context of a given design and its implementation.
4. Analyze the efficacy of a given design and implementation using empirical data.
5. Understand the implications of technology use (e.g., social media) for different identities, cultures, and communities.

KA Core:

6. Describe the internet's role in facilitating communication between citizens, governments, and each other.
7. Analyze the effects of reliance on computing in the implementation of democracy (e.g., delivery of social services, electronic voting).
8. Describe the impact of a lack of appropriate representation of people from historically minoritized populations in the computing profession (e.g., industry culture, product diversity).
9. Discuss the implications of context awareness in ubiquitous computing systems.
10. Express how access to the internet and computing technologies affect different societies.
11. Identify why/how internet access can be viewed as a human right.

SEP-Ethical-Analysis: Methods for Ethical Analysis

Ethical theories and principles are the foundations of ethical analysis because they are the viewpoints which can provide guidance along the pathway to a decision. Each theory emphasizes different assumptions and methods for determining the ethicality of a given action. It is important for students to recognize that decisions in different contexts may require different ethical theories (including combinations) to arrive at ethically acceptable outcomes, and what constitutes 'acceptable' depends on

a variety of factors such as cultural context. Applying methods for ethical analysis requires both an understanding of the underlying principles and assumptions guiding a given tool and an awareness of the social context for that decision. Traditional ethical frameworks (e.g., [11]) as provided by western philosophy can be useful, but they are not all-inclusive. Effort must be taken to include decolonial, indigenous, and historically marginalized ethical perspectives whenever possible. No theory will be universally applicable to all contexts, nor is any single ethical framework the 'best.' Engagement across various ethical schools of thought is important for students to develop the critical thinking needed in judiciously applying methods for ethical analysis of a given situation.

CS Core:

1. Avoiding fallacies and misrepresentation in argumentation
2. Ethical theories and decision-making (philosophical and social frameworks, e.g. [1])
3. Recognition of the role culture plays in our understanding, adoption, design, and use of computing technology
4. Why ethics is important in computing, and how ethics is similar to, and different from, laws and social norms

KA Core:

5. Professional checklists
6. Evaluation rubrics
7. Stakeholder analysis
8. Standpoint theory
9. Introduction to ethical frameworks (e.g., consequentialism such as utilitarianism, non-consequentialism such as duty, rights, or justice, agent-centered such as virtue or feminism, contractarianism, ethics of care) and their use for analyzing an ethical dilemma

Illustrative Learning Outcomes:

CS Core:

1. Describe how a given cultural context impacts decision making.
2. Express the use of example and analogy in ethical argument.
3. Analyze (and avoid) basic logical fallacies in an argument.
4. Analyze an argument to identify premises and conclusion.
5. Evaluate how and why ethics is so important in computing and how it relates to cultural norms, values, and law.
6. Justify a decision made on ethical grounds.

KA Core:

7. Distinguish all stakeholder positions in relation to their cultural context in a given situation.
8. Analyze the potential for introducing or perpetuating ethical debt (deferred consideration of ethical impacts or implications) in technical decisions.
9. Discuss the advantages and disadvantages of traditional ethical frameworks.

10. Analyze ethical dilemmas related to the creation and use of technology from multiple perspectives using ethical frameworks.

SEP-Professional-Ethics: Professional Ethics

Computer ethics is a branch of practical philosophy that deals with how computing professionals should make decisions regarding professional and social conduct. There are three primary influences: 1) the individual's own personal ethical code, 2) any informal or formal regulation/decree/etc. of ethical behavior existing in the workplace, applicable licensures, certifications, or laws, and 3) exposure to formal codes of ethics and ethical frameworks.

CS Core:

1. Community values and the laws by which we live
2. The nature of being a professional including care, attention, discipline, fiduciary responsibility, and mentoring
3. Keeping up to date as a computing professional in terms of familiarity, tools, skills, legal and professional frameworks as well as the ability and responsibility to self-assess and progress in the computing field
4. Professional certification, codes of ethics, conduct, and practice, such as the ACM, IEEE, AAAI, and other international societies
5. Accountability, responsibility, and liability (e.g., software correctness, reliability and safety, warranty, negligence, strict liability, ethical approaches to security vulnerability disclosures) including whether a product/service should be built, not just doing so because it is technically possible.
6. Introduction to theories describing the human creation and use of technology including instrumentalism, sociology of technological systems, disability justice, neutrality thesis, pragmatism, and decolonial models, including developing and using technology to right wrongs and do good
7. Strategies for recognizing and reporting designs, systems, software, and professional conduct (or their outcomes) that may violate law or professional codes of ethics

KA Core:

8. The role of the computing professional and professional societies in public policy
9. Maintaining awareness of consequences
10. Ethical dissent and whistleblowing
11. The relationship between regional culture and ethical dilemmas
12. Dealing with harassment and discrimination
13. Forms of professional credentialing
14. Ergonomics and healthy computing environments
15. Time-to-market and cost considerations versus quality professional standards

Illustrative Learning Outcomes:

CS Core:

1. Identify ethical issues that arise in software design, development practices, and software deployment.
2. Discuss how to address ethical issues in specific situations.
3. Express the ethical responsibility of ensuring software correctness, reliability and safety including from where this responsibility arises (e.g., ACM/IEEE/AAAI Codes of Ethics, laws and regulations, organizational policies).
4. Describe the mechanisms that typically exist for a professional to keep up to date in ethical matters.
5. Describe the strengths and weaknesses of relevant professional codes as expressions of being a professional and guides to decision-making.
6. Analyze a global computing issue, observing the role of professionals and government officials in managing this problem.

KA Core:

7. Describe ways in which professionals and professional organizations may contribute to public policy.
8. Describe the consequences of inappropriate professional behavior.
9. Be familiar with whistleblowing and have access to knowledge to guide one through an incident.
10. Identify examples of how regional culture interplays with ethical dilemmas.
11. Describe forms of harassment and discrimination and avenues of assistance.
12. Assess various forms of professional credentialing.
13. State the relationship between ergonomics in computing environments and people's health.
14. Describe issues associated with industries' push to focus on time-to-market versus enforcing quality professional standards.

SEP-IP: Intellectual Property

Intellectual property refers to a range of intangible rights of ownership in any product of human intellect, such as a software program. Laws, which vary by locality, provide different methods for protecting these rights of ownership based on their type. Ideally, intellectual property laws balance the interests of creators and users of the property. There are four types of intellectual property rights relevant to software: patents, copyrights, trade secrets, and trademarks. Moreover, property rights are often protected by user licenses. Each affords a different type of legal protection.

CS Core:

1. Intellectual property rights
2. Intangible digital intellectual property (IDIP)
3. Legal foundations for intellectual property protection
4. Common software licenses (e.g., MIT, GPL and its variants, Apache, Mozilla, Creative Commons)
5. Plagiarism and authorship

KA Core:

6. Philosophical foundations of intellectual property

7. Forms of intellectual property (e.g., copyrights, patents, trade secrets, trademarks) and the rights they protect
8. Limitations on copyright protections, including fair use and the first sale doctrine
9. Intellectual property laws and treaties that impact the enforcement of copyrights
10. Software piracy and technical methods for enforcing intellectual property rights, such as digital rights management and closed source software as a trade secret
11. Moral and legal foundations of the open-source movement
12. Systems that use others' data (e.g., large language models)

Illustrative Learning Outcomes:

CS Core:

1. Describe and critique legislation and precedent aimed at digital copyright infringements.
2. Identify contemporary examples of intangible digital intellectual property.
3. Select an appropriate software license for a given project.
4. Defend legal and ethical uses of copyrighted materials.
5. Interpret the intent and implementation of software licensing.
6. Discuss whether a use of copyrighted material is likely to be fair use.
7. Analyze the ethical issues inherent in various plagiarism detection mechanisms.
8. Identify multiple forms of plagiarism beyond verbatim copying of text or software (e.g., intentional paraphrasing, authorship misrepresentation, and improper attribution).

KA Core:

9. Discuss the philosophical bases of intellectual property in an appropriate context (e.g., country).
10. Distinguish the conflicting issues involved in securing software patents.
11. Contrast the protections and obligations of copyright, patent, trade secret, and trademarks.
12. Describe the rationale for the legal protection of intellectual property in the appropriate context (e.g., country).
13. Analyze the use of copyrighted work under the concepts of fair use and the first sale doctrine.
14. Identify the goals of the open-source movement and its impact on fields beyond computing, such as the right-to-repair movement.
15. Summarize the global nature of software piracy.
16. Criticize the use of technical measures of digital rights management (e.g., encryption, watermarking, copy restrictions, and region lockouts) from multiple stakeholder perspectives.
17. Discuss the nature of anti-circumvention laws in the context of copyright protection.

SEP-Privacy: Privacy and Civil Liberties

Electronic information sharing highlights the need to balance privacy protections with information access. The ease of digital access to many types of data – in addition to copying and distributing these data – makes privacy rights and civil liberties more complex, especially given cultural and legal differences in these areas. Complicating matters further, privacy also has interpersonal, organizational, professional/business, and governance components. In addition, the interconnected nature of online communities raises challenges for managing expectations and protections for freedom of expression in

various cultures and nations. Technology companies that provide platforms for user-generated content are under increasing pressure to perform governance tasks, potentially facing liability for their decisions.

CS Core:

1. Privacy implications of widespread data collection including but not limited to transactional databases, data warehouses, surveillance systems, cloud computing, and artificial intelligence
2. Conceptions of anonymity, pseudonymity, and identity
3. Technology-based solutions for privacy protection (e.g., end-to-end encryption and differential privacy)
4. Civil liberties, privacy rights, and cultural differences

KA Core:

5. Philosophical and legal conceptions of the nature of privacy including the right to privacy
6. Legal foundations of privacy protection in relevant jurisdictions (e.g., GDPR in the EU)
7. Privacy legislation in areas of practice (e.g., HIPAA in the US, AI Act in the EU)
8. Basic Principles of human-subjects research and principles beyond what the law requires (e.g., Belmont Report, UN Universal Declaration on Human Rights and how this relates to technology)
9. Freedom of expression and its limitations
10. User-generated content, content moderation, and liability

Illustrative Learning Outcomes:

CS Core:

1. Evaluate solutions to privacy threats in transactional databases and data warehouses.
2. Describe the role of data collection in the implementation of pervasive surveillance systems (e.g., RFID, face recognition, toll collection, mobile computing).
3. Distinguish the concepts and goals of anonymity and pseudonymity.
4. Describe the ramifications of technology-based privacy protections, including differential privacy and end-to-end encryption.
5. Identify cultural differences regarding the nature and necessity of privacy and other civil liberties.

KA Core:

6. Discuss the philosophical basis for the legal protection of personal privacy in an appropriate context (e.g., country).
7. Critique the intent, potential value, and implementation of various forms of privacy legislation and principles beyond what the law requires.
8. Identify strategies to enable appropriate freedom of expression.

SEP-Communication: Communication

Computing is an inherently collaborative and social discipline making communication an essential aspect of the profession. Much but not all of this communication occurs in a professional setting where communication styles, expectations, and norms differ from other contexts where similar technology

might be used. Both professional and informal communication conveys information to various audiences who may have different goals and needs for that information. Good communication is also necessary for transparency and trustworthiness. It is also important to note that computing professionals are not just communicators but are also listeners who must be able to hear and thoughtfully make use of feedback received from various stakeholders. Effective communication skills are not something one 'just knows' – they are developed and can be learned. Communication skills are best taught in context throughout the undergraduate curriculum.

CS Core:

1. Oral, written, and electronic team and group communication
2. Technical communication materials (e.g., source code, and documentation, tutorials, reference materials, API documentation)
3. Communicating with different stakeholders such as customers, leadership, or the public
4. Team collaboration (including tools) and conflict resolution
5. Accessibility and inclusivity requirements for addressing professional audiences
6. Cultural competence in communication including considering the impact of difference in natural language

KA Core:

7. Tradeoffs in competing factors that affect communication channels and choices
8. Communicating to solve problems or make recommendations in the workplace, such as raising ethical concerns or addressing accessibility issues

Illustrative Learning Outcomes:

CS Core:

1. Understand the importance of writing concise and accurate technical documents following well-defined standards for format and for including appropriate tables, figures, and references.
2. Analyze written technical documentation for technical accuracy, concision, lack of ambiguity, and awareness of audience.
3. Compose and deliver an audience-aware, accessible, and organized formal presentation.
4. Plan interactions (e.g., virtual, face-to-face, shared documents) with others in ways that invite inclusive participation, model respectful consideration of others' contributions, and explicitly value diversity of ideas.
5. Identify and describe qualities of effective communication (e.g., virtual, face-to-face, intragroup, shared documents).
6. Understand how to communicate effectively and appropriately as a member of a team including conflict resolution techniques.
7. Discuss ways to influence performance and results in diverse and cross-cultural teams.

KA Core:

8. Assess personal strengths and weaknesses to work remotely as part of a team drawing from diverse backgrounds and experiences.
9. Choose an appropriate way to communicate delicate ethical concerns.

SEP-Sustainability: Sustainability

Sustainability is defined by the United Nations as “development that meets the needs of the present without compromising the ability of future generations to meet their own needs.” [12] Alternatively, it is the “balance between the environment, equity and economy.” [13] As computing extends into more and more aspects of human existence, we are already seeing estimates that double-digit percentages of global electricity usage are consumed by computing activities, which unchecked will likely grow. Further, electronics contribute individually to demand for rare earth elements, mineral extraction, and countless e-waste concerns. Students should gain a background that recognizes these global and environmental costs and their potential long-term effects on the environment and local communities.

CS Core:

1. Environmental, social, and cultural impacts of implementation decisions (e.g., sustainability goals, algorithmic bias/outcomes, economic viability, and resource consumption)
2. Local/regional/global social and environmental impacts of computing systems and their use (e.g., carbon footprints, resource usage, e-waste) due to hardware (e.g., e-waste, data centers, rare element and resource utilization, recycling) and software (e.g., cloud-based services, blockchain, AI model training and use). This includes everyday use of hardware (cheap hardware replaced frequently) and software (web-browsing, email, and other services with hidden/remote computational demands).
3. Guidelines for sustainable design standards.

KA Core:

4. Systemic effects of complex computing technologies and phenomena (e.g., generative AI, data centers, social media, offshoring, remote work).
5. Pervasive computing: Information processing that has been integrated into everyday objects and activities, such as smart energy systems, social networking, and feedback systems to promote sustainable behavior, transportation, environmental monitoring, citizen science and activism.
6. How the sustainability of software systems is interdependent with social systems, including the knowledge and skills of its users, organizational processes and policies, and its societal context (e.g., market forces, government policies).

Illustrative Learning Outcomes:

CS Core:

1. Identify ways to be a sustainable practitioner in a specific area or with a specific project.
2. Assess the environmental impacts of a given project’s deployment (e.g., energy consumption, contribution to e-waste, impact of manufacturing).
3. Describe global social and environmental impacts of computer use and disposal.
4. List the sustainable effects of modern practices and activities (e.g., remote work, e-commerce, cryptocurrencies, AI models, data centers).

KA Core:

5. Describe the environmental impacts of design choices within the field of computing that relate to algorithm design, operating system design, networking design, database design, etc.
6. Analyze the social and environmental impacts of new system designs.
7. Design guidelines for sustainable IT design or deployment in areas such as smart energy systems, social networking, transportation, agriculture, supply-chain systems, environmental monitoring, and citizen activism.
8. Assess computing applications in respect to environmental issues (e.g., energy, pollution, resource usage, recycling and reuse, food management and production).

SEP-History: Computing History

History is important because it provides a mechanism for understanding why our computing systems operate the way they do, the societal contexts in which current approaches arose, and how those continue to echo through the discipline today. Not only does computing affect society but vice-versa, resulting in a complex socio-technical context that is constantly changing, requiring the perspective of history to put the present, as well as possible futures, into appropriate perspective. It also informs decisions based on successes and failures of the past including harm done and how to not repeat them. The history of computing is often taught in context with foundational concepts, such as system fundamentals and software development fundamentals. A focus should be placed on those who, due to marginalization, have not historically featured as prominently as they should.

CS Core:

1. The history of computing: hardware, software, and human/organizational.
2. The role of history in the present including within different social contexts, and the relevance of this history on the future.

KA Core:

3. Age I (Pre-digital): Ancient analog computing (Stonehenge, Antikythera mechanism, Salisbury Cathedral clock, etc.), human-calculated number tables, Euclid, Lovelace, Babbage, Gödel, Church, Turing, pre-electronic (electro-mechanical and mechanical) hardware
4. Age II (Early modern computing): ENIAC, UNIVAC, Bombes (Bletchley Park and codebreakers), computer companies (e.g., IBM), mainframes, etc.
5. Age III (PC era): PCs, modern computer hardware and software, Moore's Law
6. Age IV (Internet): Networking, internet architecture, browsers and their evolution, standards, born-on-the-internet companies, and services (e.g., Google, Amazon, Microsoft, etc.), distributed computing
7. Age V (Mobile & Cloud): Mobile computing and smartphones, cloud computing and models thereof (e.g., SaaS), remote servers, security and privacy, social media
8. Age VI (AI): Decision making systems, recommender systems, generative AI and other machine learning driven tools and technologies

Illustrative Learning Outcomes:

CS Core:

1. Understand the relevance and impact of computing history on recent events, present context, and possible future outcomes, from more than one cultural perspective.
2. Discuss how perspectives held today have been shaped by history, and that alternative perspectives exist (e.g., fears of AI replacing human workers vs AI augmenting human work, various views on Moore's Law).

KA Core:

3. Identify formative and consequential trends in the history of the computing field.
4. Identify the contributions of pioneering individuals or organizations (research labs, computer companies, government offices) in the computing field.
5. Discuss the historical context for important moments in history of computing, such as the move from vacuum tubes to transistors (TRADIC), early seminal operating systems (e.g., OS 360), Xerox PARC and the first Apple computer with a GUI, the creation of specific programming language paradigms, the first computer virus, the creation of the internet, the creation of the WWW, the dot com bubble, Y2K, the introduction of smartphones, etc.
6. Compare daily life before and after the advent of milestone developments (e.g., personal computers or the internet).

SEP-Economies: Economies of Computing

The economies of computing are important to those who develop and provide computing resources and services to others as well as society in general. They impact users of these resources and services, both professional and non-professional. Computing professionals have a duty to know the impact of these topics on their own roles and activities and how choices made will impact users and society.

KA Core:

1. Economic models: regulated and unregulated, monopolies, network effects, and open market; knowledge and attention economies
2. Pricing and deployment strategies: planned obsolescence, subscriptions, freemium, software licensing, open-source, free software, adware
3. Impacts of differences in access to computing resources, and the effect of skilled labor supply and demand on the quality of computing products
4. Automation, AI, and their effects on job markets, developers, and users
5. Ethical concerns surrounding the attention economy and other economies of computing (e.g. informed consent, data collection, use of verbose legalese in user agreements)

Illustrative Learning Outcomes:

KA Core:

1. Summarize the social effects of economic models (e.g., the knowledge and attention economies).
2. Describe the differences and similarities of competing strategies (e.g., subscription vs freemium vs free).
3. Discuss examples of digital divides.
4. Understand the effects of automation and AI on society.
5. Understand the ethical implications of computing economies that rely on attention and data.

SEP-Security: Security Policies, Laws and Computer Crimes

While security policies, laws and computer crimes are important topics, it is essential they are viewed with the foundation of other social and professional knowledge units, such as Intellectual Property, Privacy and Civil Liberties, Social Context, and Professional Ethics. Computers, the internet, and artificial intelligence, perhaps more than any other technologies, have transformed society over the past 75 years. At the same time, these technologies have contributed to unprecedented threats to privacy; new categories of computer crime and antisocial behavior; major disruptions to organizations; and the large-scale concentration of risk in information systems.

CS Core:

1. Computer crimes, legal redress for computer criminals and impact on victims and society
2. Social engineering, computing-enabled fraud, identity theft and recovery from these
3. Cyber terrorism, criminal hacking, and hacktivism
4. Malware, viruses, worms
5. Attacks on critical infrastructure such as electrical grids and pipelines
6. Non-technical fundamentals of security (e.g., human engineering, policy, confidentiality)

KA Core:

7. Benefits and challenges of existing and proposed computer crime laws
8. Security policies and the challenges of change and compliance
9. Responsibility for security throughout the computing life cycle
10. International and local laws and how they intersect

Illustrative Learning Outcomes:

CS Core:

1. List classic examples of computer crimes and social engineering incidents with societal impact.
2. Identify issues with laws that apply to computer crimes.
3. Describe the motivation and ramifications of cyber terrorism, data theft, hacktivism, ransomware, and other attacks.
4. Examine the ethical and legal issues surrounding the misuse of access and various breaches of security.
5. Discuss the professional's role in security and the tradeoffs and challenges involved.

KA Core:

6. Investigate measures that can be taken by both individuals and organizations including governments to prevent or mitigate the undesirable effects of computer crimes.
7. Design a company-wide security policy, which includes procedures for managing passwords and employee monitoring.
8. Understand how legislation from one region may affect activities in another (e.g., how EU GDPR applies globally, when EU persons are involved).

SEP-DEIA: Diversity, Equity, Inclusion, and Accessibility

Despite being a creative, often highly compensated field with myriad job (and other) opportunities, racial, gender, and many other inequities in representation are pervasive in many regions. For too many students, their first computer science course is their last. There are many factors including the legacy of systemic racism, ableism, sexism, classism, and other injustices that contribute to the lack of diverse identities within computer science, and there is no single or quick fix.

CS2023's sponsoring organizations are ACM, IEEE, and AAAI. Each of these places a high value on diversity, equity, inclusion, and accessibility; our computer science classrooms should promote and model those principles. We should welcome and seek diversity – the gamut of human differences including gender identity, ethnicity, race, politics, abilities and attributes, religion, nationality, etc.– in our classrooms, departments, and campuses. We should strive to make our classrooms, labs, and curricula accessible and to promote inclusion. We should aim for students and all community members to have a sense of belonging that comes with being respected, wanted, and valued. To achieve equity, we must allocate resources, promote fairness, and check our biases to ensure persons of all identities achieve success. Accessibility should be addressed and implemented in all computing activities and products.

Explicitly infusing diversity, equity, inclusion, and accessibility (DEIA) across the computer science curriculum demonstrates its importance for the department, institution, and our field – all of which are likely to have a DEIA statement and/or initiative(s). This emphasis on DEIA is important ethically and a bellwether issue of our time. Many professionals in computing already recognize attention to DEIA as integral to disciplinary practice. Regardless of the degree to which these values appear in any one computer science class, research suggests that a lack of attention to them results in inferior designs in addition to harm and its perpetuation. Not only does data support that diverse teams outperform homogeneous ones, but diverse teams may better prevent egregious technology failures in recent headlines such as facial recognition misuse, airbag injuries and deaths, and other well-known failures of science and computing.

CS Core:

1. How identity impacts and is impacted by computing technologies and environments (academic and professional)
2. The benefits of diverse development teams and the impacts of teams that are not diverse
3. Inclusive language and charged terminology, and why their use matters
4. Inclusive behaviors and why they matter
5. Designing and developing technology with accessibility in mind
6. How computing professionals can influence and impact diversity, equity, inclusion and accessibility, including but not only through the software they create

KA Core:

7. Experts and their practices that reflect the identities of the classroom and the world through practical DEIA principles
8. Historic marginalization due to systemic social mechanisms, technological supremacy and global infrastructure challenges to diversity, equity, inclusion, and accessibility
9. Cross-cultural differences in, and needs for, diversity, equity, inclusion, and accessibility

Illustrative Learning Outcomes:

CS Core:

1. Define and distinguish equity, equality, diversity, inclusion, and accessibility.
2. Identify language, practices, and behaviors that may make someone feel included in a workplace and/or a team, and why is it relevant. Avoid charged terminology - see *Words Matter* (www.acm.org/diversity-inclusion/words-matter) – this includes identifying and accommodating users who are often excluded without thought and not considered at all.
3. Evaluate the accessibility of your classroom or lab. Evaluate the accessibility of your webpage. (See www.w3.org/WAI and www.w3.org/TR/WCAG21).
4. Demonstrate collegiality and respect when working with team members who do not share your identity. *It is not enough to merely assign team projects. Faculty should prepare students for teamwork and monitor, mentor, and assess the effectiveness of their student teams throughout a project.*
5. Compare the demographics of your institution's computer science and STEM majors to the overall institutional demographics. If they differ, identify factors that contribute to inequitable access, engagement, and achievement in computer science among marginalized groups. If they do not, assess why not.
6. Identify developers' assumptions and values embedded in hardware and software design, especially as they pertain to usability by diverse populations.
7. Identify examples of the benefits that diverse teams can bring to software products, and how a lack of diversity has costs.

KA Core:

8. Analyze the work of experts who reflect the identities of the classroom and the world.
9. Assess the impact of power and privilege in the computing profession as it relates to culture, industry, products, and society.
10. Develop examples of systemic changes that could positively address diversity, equity, inclusion, and accessibility in a familiar context (i.e., in an introductory computing course) and an unfamiliar context and when these might be different, or the same.
11. Compare the demographics of your institution to the overall community demographics. If they differ, identify factors that contribute to inequitable access, engagement, and achievement among marginalized groups. If they do not, assess why not.

Professional Dispositions

- **Critically Self-reflective:** Students should be able to inspect their own actions, thoughts, biases, privileges, and motives to discover places where professional activity is not up to current standards. They must strive to understand both conscious and unconscious biases and continuously work to counteract them.
- **Responsive:** Students must quickly and accurately respond to changes in the field and adapt in a professional manner, such as shifting from in-person office work to remote work at home. These shifts require rethinking one's entire approach to what is considered "professional".